SNA PROJECT 2024

Aim: demonstrate the application of the techniques we studied in class

Where to find network?

See repositories on ELO (Snap, Koblenz, Kaggle…)

Wikipedia script

Your own network (find good API or scrape data)

Type of Analyses

Network descriptive statistics

- Number of nodes, links
- Density
- Diameter
- Avg. Shortest path
- Cluster Coefficient
- Number of components / Size of the giant component
- Degree Distribution
  - What shape is that?
- What model better describes your network (Random? Small World? Barabasi?)
- Reciprocity
- Transitivity and triad census
- Analysis of triads (FANMOD)

Cluster Analysis

- Modularity, hierarchical clustering
- After clustering try to label each cluster

Centrality measures

- Degree
- Betweenness
- Closeness
- PageRank

Resilience Analysis

- Removing nodes or edges
- Random of with a strategy
- Check the size of the giant connected component
- Check the length of the shortest path

Diffusion over a network

- Simulate the spread of “something” on the network
- Difference strategies for contagion (simple, threshold, probabilistic, complex)
- Strategies to stop the diffusion

Ego Network Analysis

Advanced analysis

- Homophily
- Assortativity (degree or any other variable)
- Regression models between network metrics and other external indicators/metadata

Ideas for projects

Power law in big network	Build or collect big networks and check if the degree distribution is a power. You don't have to build the network but collect the number of in-links of a big sample of nodes. Find if the degree distribution is fitting a power law and with which exponent	
Are real network small worlds?		
Resilience of a Network	See example about German rail network	
Contagion over a network	See example of past project and the python script shown in class	
Ego network	Create a set of ego networks. For each "ego" you should collect some metadata. Compute ego networks metrics (like density, efficienvy, cluster coefficient, size) Check if the network metrics of each ego are correlated to its metadata (using correlation or regression analysis). Example: Collect a network of singer. For instance, use the Wikipedia script with a list of singer Collect singer metadata like the number of sales, songs streamed, grammies award won and check if those numbers are correlated with the ego network metrics for the same singer. You can also perform a regression analysis where the target variable is (for instance) is the number of song streamed from that singer and the predictors are the ego network metrics for that singer.	
Relationship about network metrics and metadata of nodes	Similar project as the previous one, but this time you have the full network instead of the EGO networks. You can extract from the full network some node statistics (like centrality, clustering coefficient) , then collect metadata about the node and check if the network features are correlated to the metadata. Collect or download a network of movies, collected the rating that each movie received, and check if there is a correlation between the rating and the centrality of the node in the network. Are the most central movies the ones with higher rating?	